

*MA.7.NSO.2.1***Benchmark**

MA.7.NSO.2.1 Solve mathematical problems using multi-step order of operations with rational numbers including grouping symbols, whole-number exponents and absolute value.

Benchmark Clarifications:

Clarification 1: Multi-step expressions are limited to 6 or fewer steps.

Connecting Benchmarks/Horizontal Alignment

- MA.6.NSO.1.4
- MA.6.AR.1.3
- MA.7.NSO.1.1
- MA.7.AR.1
- MA.7.GR.1.1, MA.7.GR.1.2

Terms from the K-12 Glossary

- Absolute Value
- Exponent
- Order of Operations
- Rational Number

Vertical Alignment**Previous Benchmarks**

- MA.5.AR.2.2, MA.5.AR.2.3

Next Benchmarks

- MA.7.NSO.1
- MA.8.NSO.1.5, MA.8.NSO.1.7

Purpose and Instructional Strategies

In previous courses, students used the order of operations to evaluate multi-step numerical expressions including parentheses. In grade 6 accelerated, students evaluate algebraic expressions using substitution and order of operations with integers, including use of absolute value and natural number exponents. Students also solve multi-step order of operations with rational numbers including grouping symbols, whole-number exponents and absolute value. In future courses, students will solve problems involving order of operations involving radicals.

- Number sense and properties of operations should be emphasized during instruction as this benchmark is the completion of operations with rational numbers.
- Remind students that subtraction is addition of an opposite and division is multiplication by a reciprocal when working with order of operations (*MTR.3.1*).
- Avoid mnemonics, such as PEMDAS, that do not account for other grouping symbols and do not exercise proper number sense that allows for calculating accurately in a different order.
- Instruction includes the use of technology to help emphasize the proper use of grouping symbols for order of operations.
- Students should have experience using technology with decimals and fractions as they occur in the real world. This experience will help to prepare students working with irrational numbers in grade 8.

Common Misconceptions or Errors

- Students may confuse when parentheses are used for grouping or multiplication.
- Some students may incorrectly apply the order of operations. In order to support students in moving beyond this misconception, be sure to review operations with rational numbers and order of operations.

Strategies to Support Tiered Instruction

- Instruction includes the use of colors to highlight each step of the process used to evaluate an expression.
- Teacher co-creates a graphic organizer for different grouping symbols and provides examples when the grouping symbols indicate operator symbols.
 - For example, students can be given the expressions below and discuss similarities and differences.

$$\begin{array}{ccc} \left(\frac{4}{6} + 9\right) + 87 & \left(\frac{4}{6} + 9\right) 87 & \left(\frac{4}{6} + 9\right) - 87 \\ \left(\frac{4}{6} + 9\right) (+87) & \left(\frac{4}{6} + 9\right) (87) & \left(\frac{4}{6} + 9\right) (-87) \end{array}$$

- Instruction includes reviewing operations with rational numbers and order of operations.

Instructional Tasks

Instructional Task 1 (MTR.4.1, MTR.5.1)

Part A. Using the integers -6 to 6 at most once, fill in the boxes to create an expression with the lowest value.

$$\frac{\boxed{}}{\boxed{}} \left(\boxed{} - \frac{\boxed{}}{\boxed{}} \parallel \boxed{} - \boxed{} \right)$$

Part B. Compare your value with those in your group. Who has the lowest value? Explain why this value was less than the others.

Instructional Task 2 (MTR.3.1, MTR.4.1)

Part A. Evaluate the expression $\left(\frac{\left|\frac{3}{4}(24) - 21\right|}{2}\right)^4$.

Part B. Compare your strategy with a partner.

Instructional Items

Instructional Item 1

What is the value of the expression $\frac{(12 - |8 - 5|)^3}{36}$?

Instructional Item 2

What is the value of the expression $\frac{1}{2}(3^2 - 4) + \left|7 - \frac{1}{6}\right|$?

Instructional Item 3

Evaluate the expression $18 - 3(4.12 + 7.6 \div 2)$.